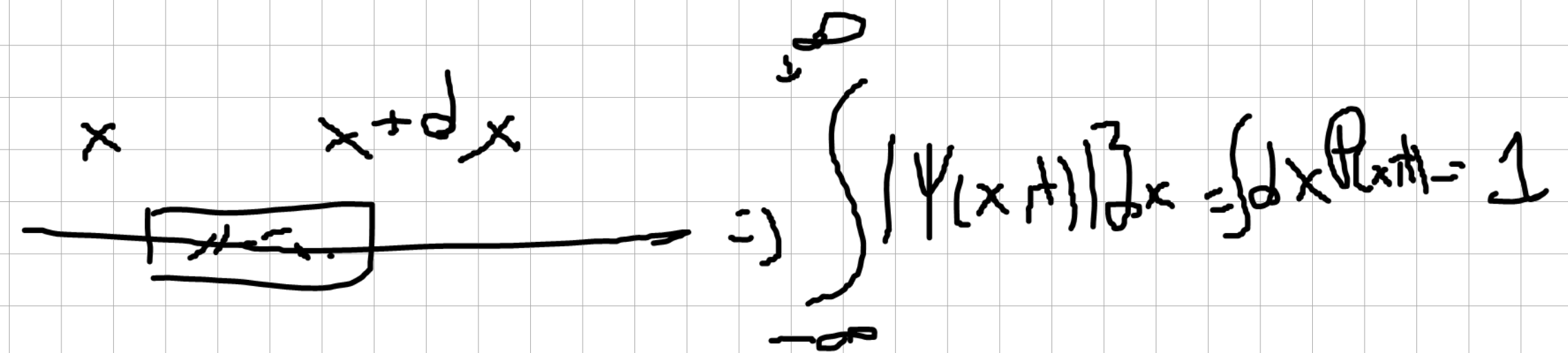


Quantum Mekaniği: Bölüm 3

Schr. Denk ve Olasılık Yorumu

klasik

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} \Rightarrow |\Psi(x,t)|^2 dx = P(x,t) dx$$



$$\int_{-\infty}^{\infty} |\Psi(x,t)|^2 dx = \int_{-\infty}^{\infty} P(x,t) dx = 1$$

$$P(x,t) = |\Psi(x,t)|^2 = \Psi(x,t) \Psi^*(x,t) \text{ max kin teor}$$

$$\frac{dP}{dt} = \Psi^* \frac{d\Psi}{dt} + \Psi \frac{d\Psi^*}{dt} \Rightarrow \frac{d\Psi}{dt} = -\frac{\hbar^2}{2im\hbar} \frac{\partial^2 \Psi}{\partial x^2} \quad \frac{d\Psi^*}{dt} = +\frac{\hbar^2}{2im\hbar} \frac{\partial^2 \Psi^*}{\partial x^2}$$

$$\Rightarrow \frac{dP}{dt} = \Psi^* \left(-\frac{\hbar^2}{2im\hbar} \frac{\partial^2 \Psi}{\partial x^2} \right) + \Psi \left(\frac{\hbar^2}{2im\hbar} \frac{\partial^2 \Psi^*}{\partial x^2} \right)$$

$$\Rightarrow \frac{dP}{dt} = \frac{\hbar^2}{2im\hbar} \left(-\Psi^* \frac{\partial^2 \Psi}{\partial x^2} + \Psi \frac{\partial^2 \Psi^*}{\partial x^2} \right)$$

$$\frac{d}{dx} \left(\Psi \frac{d\Psi^*}{dx} \right) = \frac{d\Psi}{dx} \frac{d\Psi^*}{dx} + \Psi \frac{\partial^2 \Psi^*}{\partial x^2}$$

$$\frac{d}{dx} \left(\Psi^* \frac{d\Psi}{dx} \right) = \frac{d\Psi^*}{dx} \frac{d\Psi}{dx} + \Psi^* \frac{\partial^2 \Psi}{\partial x^2}$$

$$\frac{dP}{dt} = \frac{\hbar^2}{2im\hbar} \left(\frac{d}{dx} \left(\Psi \frac{d\Psi^*}{dx} - \Psi^* \frac{d\Psi}{dx} \right) \right) \Rightarrow \frac{dP}{dt} - \frac{d}{dx} \left(\frac{\hbar^2}{2im\hbar} \left(\Psi \frac{d\Psi^*}{dx} - \Psi^* \frac{d\Psi}{dx} \right) \right) = 0$$

$$\frac{dP}{dt} + \frac{d}{dx} \underbrace{\left(\frac{\hbar^2}{2im\hbar} \left(\Psi^* \frac{d\Psi}{dx} - \Psi \frac{d\Psi^*}{dx} \right) \right)}_{\nabla J} = 0$$

Korunum Denk

$\vec{p}$ = momentum

$$i\hbar \frac{\partial \Psi(r,t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi + V(r) \Psi$$

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V(x) \Psi \quad \Psi = \Psi(x,t)$$

$$\Rightarrow \frac{d\Psi}{dt} = -\frac{\hbar^2}{2im\hbar} \frac{\partial^2 \Psi}{\partial x^2} + \frac{V}{i\hbar} \Psi \quad \frac{d\Psi^*}{dt} = +\frac{\hbar^2}{2im\hbar} \frac{\partial^2 \Psi^*}{\partial x^2} - \frac{V^*}{i\hbar} \Psi^* \Rightarrow \frac{dP}{dt} = \Psi \frac{d\Psi^*}{dt} + \Psi^* \frac{d\Psi}{dt}$$

$$\Rightarrow \frac{dP}{dt} = \Psi \frac{\hbar^2}{2im\hbar} \frac{\partial^2 \Psi^*}{\partial x^2} - \Psi \frac{V^*}{i\hbar} \Psi^* - \Psi^* \frac{\hbar^2}{2im\hbar} \frac{\partial^2 \Psi}{\partial x^2} + \Psi^* \frac{V}{i\hbar} \Psi$$

$$\Rightarrow \frac{dP}{dt} = -\frac{\hbar}{2im} \left[\Psi^* \frac{\partial^2 \Psi}{\partial x^2} - \Psi \frac{\partial^2 \Psi^*}{\partial x^2} \right] + \underbrace{\frac{\Psi^* V \Psi - \Psi V^* \Psi^*}{i\hbar}}_{0} \quad V = a+ib \quad b=0 \Rightarrow V=V^*$$

$$b \neq 0 \Rightarrow \frac{dP}{dt} + \frac{d}{dx} J = \frac{2im}{\hbar} V$$

$V^* = -b$
 $V = b$
 $2b$

$$p = \hbar k \quad E = \frac{p^2}{2m} = \hbar \omega \quad \lambda = \frac{2\pi}{k} \quad i \frac{p x}{\hbar} - i \frac{E t}{\hbar}$$

$$\int dk g(k) e^{i k x - i \omega(k) t} \Rightarrow \frac{1}{2\pi \hbar} \int dp g(p) e^{i p x / \hbar - i E(p) t / \hbar} = \Psi$$

$$\frac{d\Psi}{dp} = \frac{i x}{\hbar} \Psi \Rightarrow \frac{d\Psi}{\Psi} = \frac{i x}{\hbar} dp \Rightarrow \frac{\hbar d\Psi}{i x \Psi} = dp \Rightarrow \frac{\hbar}{i} \frac{d\Psi}{dx} = \int \Psi dp$$

$$\hat{p} = \frac{\hbar}{i} \frac{d}{dx} \Rightarrow p = \frac{\hbar}{i} \nabla$$

Momentum uzayında Dalg Fonk

$$\Psi(x) = \frac{1}{\sqrt{2\pi \hbar}} \int_{-\infty}^{+\infty} dp \phi(p) e^{i p x / \hbar} \quad p = \hbar k$$

$$\Rightarrow \Psi(x) = \frac{1}{\sqrt{2\pi \hbar}} \int d(\hbar k) \phi(\hbar k) e^{i k x}$$

Ters Dönüşüm

$$\phi(p) = \frac{1}{\sqrt{2\pi \hbar}} \int dx \Psi(x) e^{-i p x / \hbar}$$

$$\Rightarrow \int_{-\infty}^{+\infty} dx |\Psi(x)|^2 = 1 \Rightarrow \int_{-\infty}^{+\infty} dx \Psi^* \Psi = \frac{1}{2\pi \hbar} \int dx \underbrace{\int dp \phi^* e^{-i p x / \hbar}}_{\Psi^*} \cdot \underbrace{\int dp' \phi(p') e^{i p' x / \hbar}}_{\Psi} \frac{1}{\hbar}$$

$$\Rightarrow \delta(p - p') = \frac{1}{2\pi \hbar} \int_{-\infty}^{+\infty} dx e^{i(p - p')x / \hbar}$$

	<u>Grizisel mom</u>	<u>Konum</u>
<u>Reel Uzay</u>	$p = \frac{\hbar}{i} \frac{d}{dx}$	x
<u>Mom Uzay</u>	p	$x = \frac{\hbar}{i} \frac{d}{dp}$

Momentum uzayında beklenen değer p nin nedir?

$$\langle p \rangle = ? \Rightarrow \int dp \phi^*(p) p \phi(p) = p$$

$$\langle x \rangle = ? \Rightarrow \int dp \phi^*(p) x \phi(p) = i \hbar \frac{d}{dp}$$

